



Department of Computer Science

CPSI 27603: Programming II

Section 991: in WEB

Summer 2026

Programming II (991)

Instructor Info

Name: Nicholas Seward

Preferred Pronouns: he/him

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Office Location: EIT 575

Office Hours: Summer Random Hours

I am available for in-person or virtual office hours upon request. Just contact me via email or Discord to set up a time. Use the Zoom Class link for virtual meetings.

Course Description

Credit Hour(s): 3

Description:

Advanced programming concepts including abstract data types, details of object-oriented concepts such as encapsulation and polymorphism in a current object-oriented language as well as an introduction to object-oriented analysis and design, software design patterns, and software-engineering topics. The laboratory provides ample opportunity to deepen

programming skills and practical experiences in current software-development technologies.

Prerequisites: CPSC 17503 or equivalent

Course Learning Objectives

Upon successful completion of this course, students will be able to:

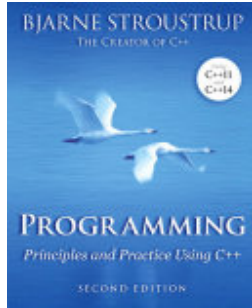
- Research a computing problem and choose solutions from non-computing disciplines. (PI-1b)
- Given a set of requirements, produce a design for a software-based solution. (PI-2a)
- Given a set of requirements, implement a program that meets the requirements. (PI-2b)
- Given user requirements, develop a test plan to evaluate a computer-based solution and determine its correctness. (PI-2c)
- Divide a complex computing problem into solvable subcomponents with appropriate data structures and algorithms. (PI-1a)

Course Materials

Title: <https://txtbook.dev/NicholasSeward/CppBook>

Required vs Recommended: Required

<https://txtbook.dev/NicholasSeward/CppBook>

**Title: Programming****Subtitle:** Principles and Practice Using C++**ISBN:** 9780321992789**Authors:** Bjarne Stroustrup**Publisher:** Pearson Education**Publication Date:** 2014-01-01**Required vs Recommended:** Optional

Written by the creator of C++, this book is for those who want to dive deep into their understanding of C++. Highly recommended if you are serious about programming.

Grading Scale

Your semester grade for this course will be determined based on the following scale:

A = 90-100

B = 80-89

C = 70-79

D = 60-69

F = less than 60

A = excellent work; B = above average work; C = average work; D = below average; F = failing

Grading Criteria

Category	Percent of Total Grade
Quiz	10%
Tutorial	10%
Practice	10%
Participation	10%
Project	30%
Midterm	10%
Final	20%

Learning Activities

Course content will be delivered through recorded lectures and assigned readings.

The primary form of assessment will be programming projects. In addition, there will be quizzes, practice programming exercises, and tutorials interspersed between the main assignments.

There will be two major assessments: a midterm and a final exam.

University Policies

Syllabus, Office Hours, and Regular and Substantive Contact

- <https://ualr.edu/policy/facstaff/course-syllabus/>

Students with Disabilities

- <https://ualr.edu/policy/facstaff/course-syllabus/>

If you are a student with a disability, or if you simply want to discuss resources that might help you learn more effectively, you can contact the Disability Resource Center at any time. You are welcome to drop in or call 501-916-3143 to make an appointment. Their staff will work with you to discuss accommodations and identify resources on campus or in the community that might be helpful for you. Accommodations are established through an interactive process. For more information, visit the DRC website or review the steps to request accommodations. We want you to know you are valued, welcome, and wanted at UA Little Rock.

Non-Discrimination Policy

- <https://ualr.edu/policy/home/admin/non-discrimination/>

Title IX and Harassment

- <https://ualr.edu/titleix/titleix/title-ix-the-basics/>

Inclement Weather Policy

- <https://ualr.edu/policy/admin/weather/>

Academic Integrity

- <https://ualr.edu/deanofstudents/academic-integrity/>

Disclosure of Instances of Sexual Misconduct

- <https://ualr.edu/policy/home/facstaff/title-ix/>

Withdrawal

- <https://ualr.edu/policy/home/student/withdrawal-from-ualr/>

Add/Drop

- <https://ualr.edu/records/2511-2/>

Incomplete

- <https://ualr.edu/records/grades/>

Grade Policies

- <https://ualr.edu/policy/home/facstaff/grades-and-grading-systems/>

Course Policies

Class Attendance & Participation

Participation is evaluated periodically through:

- Real-time interaction during synchronous class sessions.
- Contributions to online discussion boards, and
- Occasional in-class or short response assignments.

Assignment Grading Scale

- 100% – No issues
- 90% – Minor issue(s) that can be corrected quickly
- 80% – Several small issues present
- 70% – Multiple issues, but sufficient understanding demonstrated; resubmission not required
- 0% – Significant issues indicating a breakdown in understanding; the student would benefit from resubmitting the assignment

Project Grading Policy

- 90–100% – Project exceeds the minimum requirements
- 90% – Project meets all minimum requirements with no bugs
- 80% – Project meets all minimum requirements with minor bugs
- 50% – Project meets some requirements, but is incomplete
- 0% – Project is missing most required components

Late Policy

All work except exams may be submitted late.

- Due to summer craziness, you will have 1 week after a due date before a -20% penalty is assessed.

Assignment Resubmission Policy

Students may resubmit assignments an unlimited number of times. However, the late submission policy still applies. If a resubmission is made a week after the due date, it will have a -29% penalty.

Extension Policy

Extensions are not granted. However, late work is always accepted. If you experience extenuating circumstances, please inform me as a form of documentation. I will take these situations into consideration at the end of the semester. In such cases, I may adjust your grade if it appears that you

faced an isolated challenge during the semester but were able to recover successfully.

Incompletes

Unexpected events—such as work, illness, injury, or a death in the family—can affect your ability to complete the course. In such cases, we can discuss the option of an incomplete, allowing you extra time past the course's end to complete the work. This often involves completing slightly different assignments than those given to other students.

Incompletes require official paperwork and are granted at the sole discretion of the instructor.

Makeup Policy

The instructor must be informed before the midterm or final exam date if a different date is required.

Academic Honesty

Plagiarism on any assignment will at a minimum result in 0 points for the assignment. We reserve the right to pursue further disciplinary action if appropriate (e.g. any student caught cheating on an assignment/assessment will receive an "F" for the course, and we may pursue action with the Committee on Academic Integrity). Plagiarism includes copying someone else's work and claiming it as your own, or collaborating excessively with another person or persons and claiming the work as solely your own. It is strongly recommended that students maintain a record of the preparation of their major assignments. For more information on academic offenses, please refer to the following websites:

Academic Integrity and Grievance Policy

<https://ualr.edu/deanofstudents/section-vii-administration/academic-integrity-grievance-policy/> and Academic Offenses:

<https://ualr.edu/deanofstudents/section-vii-administration/academic-offenses/>

AI Usage Policy

Use of AI in this course is encouraged and permitted under the following constraints:

- Use of the AI on any work product submitted is acknowledged.
- The work submitted can be reasonable and ethically claimed as you own.
- You are fully responsible for all the content of work submitted.

Failure to adhere to these constraints may be treated as academic misconduct.

Helpful University Resources

Blackboard Student Support

- <https://ualr.edu/blackboard/contact/>

Bookstore

- <https://ualr.bnccollege.com/>

Care Team

- <https://ualr.edu/studentssuccess/care-team/>

Career Center

- <https://ualr.edu/careers/>

Child Care Connections

- <https://ualr.edu/studentssuccess/child-care-connections/>

Disabilities Resource Center

- <https://ualr.edu/disability/>

Communication Skill Center (CSC)

- <https://ualr.edu/appliedcomm/communication-skill-center-csc/>

Counseling

- <https://ualr.edu/counseling/>

Handshake

- <https://ualr.joinhandshake.com/login>

Health Services

- <https://ualr.edu/health/>

Information Technology Services (ITS)

- <https://ualr.edu/itservices/welcome/>

Mathematics Assistance Center

- <https://ualr.edu/mathematics/mathematics-assistance-center/>

Military Student Success Center

- <https://ualr.edu/military/>

Office of Student Retention Initiatives (SRI)

- <https://ualr.edu/studentsuccess/academic-resources/coaching-and-student-support/>

Student Support Services (SSS)

- <https://ualr.edu/studentssupport/>

Trojan Tutoring (i.e. introductory-level courses only) via University Learning Commons

- <https://ualr.edu/studentsuccess/academic-resources/>

UA Little Rock Trojan Food Pantry

- <https://ualr.edu/foodpantry/>

University Writing Center (UWC)

- <https://ualr.edu/writingcenter/>

Ottenheimer Library

- <https://ualr.edu/library/>

Tentative Course Schedule

C++ Fundamentals Review (Module 1)

- Basic syntax, variables, data types
- Input/output operations, control structures
- Compilation process and IDEs

Vectors and Random Numbers (Module 2)

- Introduction to `std::vector`
- Dynamic resizing and memory layout
- Random number generation and loops

Module 3: Functions and File I/O (Module 3)

- Function declaration, definition, and calls
- Reference vs value parameters, function overloading
- File input/output with `fstream`
- **PROJECT 1 DUE**

Unit Testing and Software Development (Module 4)

- Writing unit tests
- Test-driven development and assertions
- Clean coding practices and organization

Templates and STL Containers (Module 5)

- Function templates
- STL containers: `vector`, `map`, `set`, `unordered_map`
- Container iteration and access

Iterators, Algorithms, and Lambdas (Module 6)

- Iterator types and STL algorithms
- Lambda functions and functional programming
- **PROJECT 2 DUE**

MIDTERM EXAM

- Comprehensive exam covering Modules 1-7 material

Classes and Inheritance (Module 7)

- Class declaration and definition
- Member variables, methods, constructors, destructors
- Inheritance and protected access

GUI Programming (Module 8)

- Lightweight GUI via CLI or SDL2/ImGui
- Event loop basics and GUI inputs
- Connecting model and view

Operator Overloading and Polymorphism (Module 9)

- Overloading operators (<<, ==, +, etc.)
- Virtual functions and dynamic dispatch
- Abstract classes and interfaces
- **PROJECT 3 DUE**

Pointers and Smart Pointers (Module 10)

- Raw pointers, new/delete
- Smart pointers: unique_ptr, shared_ptr, weak_ptr
- Ownership semantics and RAI

Linked Lists and Trees (Module 11)

- Building singly linked lists with Node*
- Binary Search Trees: insert, search, traverse
- Manual memory management and cleanup

Design Patterns (Module 12)

- Strategy, Observer, Factory, Decorator patterns
- When and why to use design patterns
- Pattern implementation examples
- **PROJECT 4 DUE**

Graphs and Advanced Data Structures (Module 13)

- Graph representations: adjacency list vs matrix
- Depth-First Search (DFS) and Breadth-First Search (BFS)
- Optional: shortest path or topological sort

FINAL EXAM

- Comprehensive final exam covering entire semester